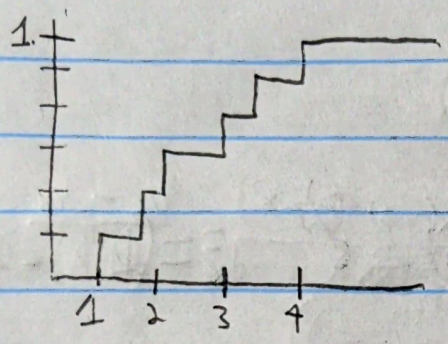
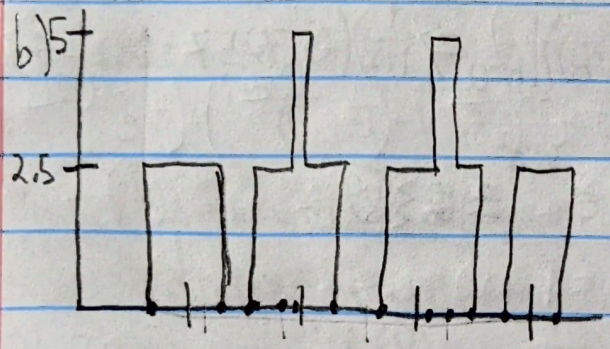
T.J. Smith KDE haz 9xm

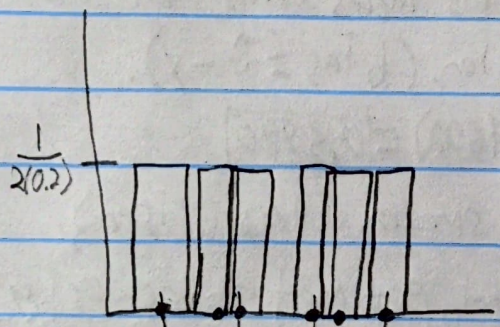
1. a)



b)



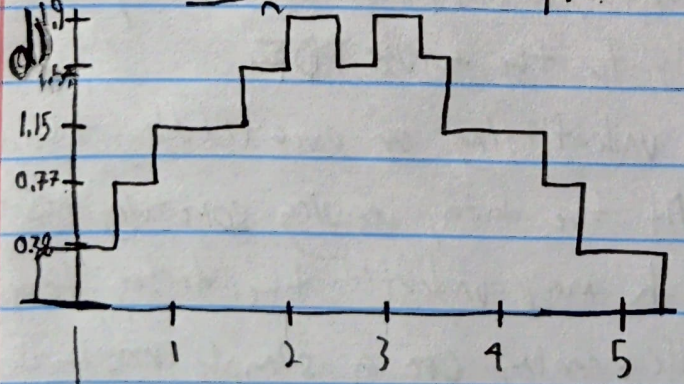
$$UKDE(x) = \begin{cases} 2.5 & 0.8 \le x \le 1.2 \\ 2.5 & 1.55 < x \le 1.80 \\ .5 & 1.80 \le x \le 1.95 \\ 2.5 & 1.95 < x \le 2.20 \\ 2.5 & 2.80 \le x \le 3.05 \\ 5 & 3.05 < x \le 3.20 \\ 2.5 & 3.20 < x \le 3.45 \\ 2.5 & 3.80 < x \le 4.20 \\ 0 & \text{all other } x \end{cases}$$

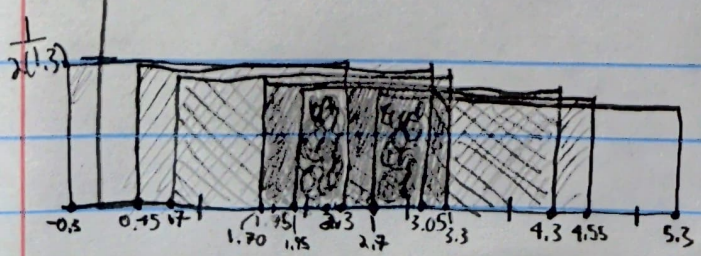


c) $h = 1.84 \cdot \sigma \cdot n^{-0.2} = 1.84 \cdot 1.07 \cdot 6^{-0.2} = 1.299$

$\sigma = \sqrt{\frac{\sum (x-\bar{x})^2}{n}} \quad \bar{x} = 2.5 \Rightarrow 1.075$

d)





e) They have peaks at the same place, but b has valleys or no value at points & does not. D looks almost normal, but b looks like a piece-wise function.

2. a) $\hat{f}_h(x) = \frac{1}{nh}\sum K\left(\frac{x-x_i}{h}\right) =$

$\hat{f}_h(0) = \frac{1}{5(0.5)}\left(\frac{1}{2}+\frac{1}{2}+\frac{1}{2}\right) = 0.50$

b)

$$E(\hat{f}_h(0)) = \int_{-h}^{h} \frac{1}{2h} f(x)\,dx = \frac{1}{2h}\int_{-h}^{h} \frac{1}{\sqrt{2\pi}} e^{\left(-\frac{1}{2}x^2\right)} dx = \left(\Phi(h) - \Phi(-h)\right)\frac{1}{2h}$$

c) $E(\hat{f}_h(x)) = \frac{1}{2h}\left(\Phi(h) - \Phi(-h)\right)\Big|_{h=0.1} \Rightarrow \frac{1}{1.2}(0.7257 - 0.2743) = \boxed{0.376}$

c) $\hat{f}_h(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}\Big|_{x=0} \Rightarrow \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}0^2} = \boxed{0.3989}$

d) bias do not equal. This means that the KDE is inherently biased by definition. (bias $= \hat{y} - y$).

e) $E(\hat{f}_{0.1}(0)) = \frac{1}{0.2}(0.5398 - 0.4602) = \boxed{0.3980}$

The difference is much smaller (less biased).

f) The KDE is a consistent estimator because given the same inputs you will find the same output. However, it is biased in that until $n = \infty$ and $h = 0$, you will not converge fully to the true PDF.

g) You want to avoid underfitting or overfitting the data. A small h may overfit the data, leaving something resembling a rug-plot. A large h may underfit the data, leaving a bell curve with little meaning or a simple rectangle.

3. a) $\hat{f}_{1.5}(6) = \frac{1}{2nh} \text{II}\{|x_i - x| \le h\} = \frac{1}{2(5)(1.5)} \text{II}\{|x_i - 6| \le h\}$

$= \frac{1}{2(5)(1.5)}(3) = \boxed{0.20}$

$\hat{f}_{1.5}(6.5) = \frac{1}{15}(4) = \boxed{0.267}$

$\hat{f}_{1.5}(7) = \frac{1}{15}(3) = \boxed{0.20}$

$\hat{f}_{1.5}(8) = \frac{1}{15}(3) = \boxed{0.20}$

$\hat{f}_{1.5}(8.5) = \frac{1}{15}(4) = \boxed{0.267}$

b) $\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^{n} \frac{1}{\sqrt{2\pi}} e^{\wedge}\left(-\frac{1}{2}\left(\frac{x_i - x}{h}\right)^2\right)$

$\hat{f}_{1.5}(6) = \frac{1}{(5)(1.5)}\left[\frac{1}{\sqrt{2\pi}} e^{\wedge}\left(-\frac{1}{2}\left(\frac{5-6}{1.5}\right)^2\right) + \frac{1}{\sqrt{2\pi}} e^{\wedge}\left(-\frac{1}{2}\left(\frac{7-6}{1.5}\right)^2\right) + \frac{1}{\sqrt{2\pi}} e^{\wedge}\left(-\frac{1}{2}\left(\frac{7-6}{1.5}\right)^2\right) + \frac{1}{\sqrt{2\pi}} e^{\wedge}\left(-\frac{1}{2}\left(\frac{8-6}{1.5}\right)^2\right)\right.$

$\left. + \frac{1}{\sqrt{2\pi}} e^{\wedge}\left(-\frac{1}{2}\left(\frac{10-6}{1.5}\right)^2\right)\right] = \frac{1}{7.5}\left(\frac{1}{\sqrt{2\pi}}(0.8007 + 0.8007 + 0.8007 + 0.4111 + 0.0286)\right)$

$= \boxed{0.1512}$

$\hat{f}_{1.5}(6.5) = \frac{1}{7.5}\left(\frac{1}{\sqrt{2\pi}}(0.6065 + 0.9460 + 0.9460 + 0.6065 + 0.0657)\right.$

$= \boxed{0.1687}$

$\hat{f}_{1.5}(7) = \frac{1}{7.5}\left(\frac{1}{\sqrt{2\pi}}(0.4111 + 1 + 1 + 0.8007 + 0.1353)\right) = \boxed{0.1780}$

$\hat{f}_{1.5}(8) = \frac{1}{7.5}\left(\frac{1}{\sqrt{2\pi}}(0.1353 + 0.8007 + 0.8007 + 1 + 0.4111)\right) = \boxed{0.1674}$

$\hat{f}_{1.5}(8.5) = \frac{1}{7.5}\left(\frac{1}{\sqrt{2\pi}}(0.0657 + 0.6065 + 0.6065 + 0.9460 + 0.6065)\right) = \boxed{0.1506}$

c) The UKDE jumps between 6.0 and 6.5, and 8.0 and 8.5. It does this because we assign a step-function to each point so when two points fall within one-another in the bandwidth, they both contribute their full density. But with GKDE, we assign a normal curve to each point, so when they fall within one-another's bandwidth they contribute only the portion of their normal curve that falls within the bandwidth.

d) Different computational capabilities and use cases will prefer different types of KDE's. Uniform KDE's use less overhead and work well when strict boundaries must be enforced. Gaussian KDE's are continuous and may do a better job capturing variability in certain data-sets.

Problem 4

In [1]:
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib as plt
from nPest.np import ecdf ,kde
```

In [2]:
```
df = pd.read_csv('data/heart_failure_clinical_records_dataset.csv')
df.head(3).T
```

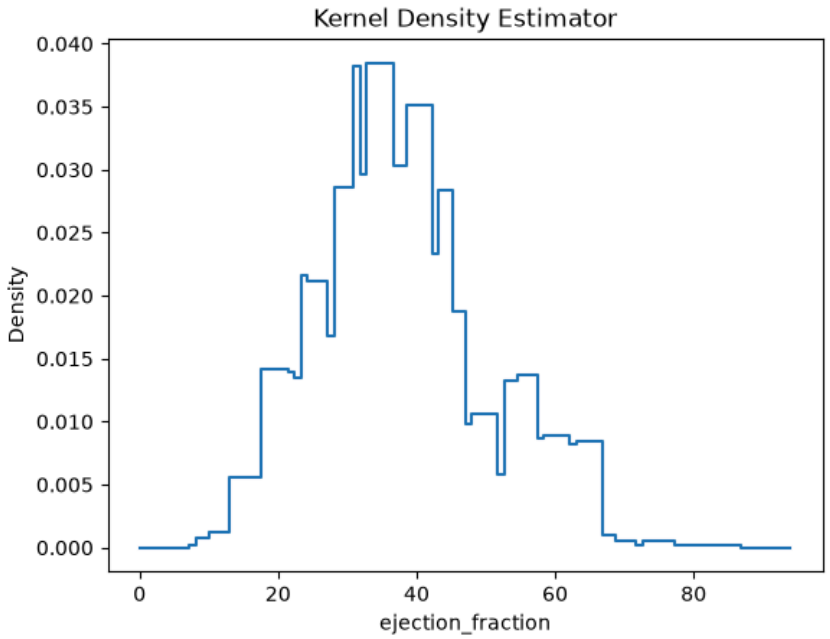
Out[2]:

	0	1	2
age	75.0	55.00	65.0
anaemia	0.0	0.00	0.0
creatinine_phosphokinase	582.0	7861.00	146.0
diabetes	0.0	0.00	0.0
ejection_fraction	20.0	38.00	20.0
high_blood_pressure	1.0	0.00	0.0
platelets	265000.0	263358.03	162000.0
serum_creatinine	1.9	1.10	1.3
serum_sodium	130.0	136.00	129.0
sex	1.0	1.00	1.0
smoking	0.0	0.00	1.0
time	4.0	6.00	7.0
DEATH_EVENT	1.0	1.00	1.0

A.

In [35]:
```
model = kde()
model.fit(x)
model.plot()
print(f'Bandwidth: {model.h:.2f}')
```

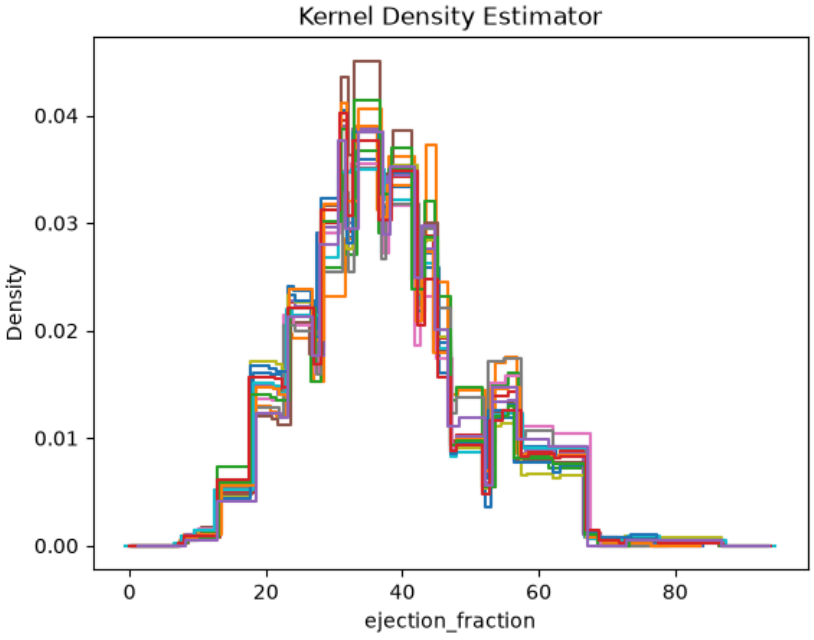
```
Bandwidth: 6.95
```



B.

In [34]:
```
model = kde()
hsum = 0
for i in range(15):
    df_i = df.sample(frac=1.0,replace=True)
    model.fit(df_i['ejection_fraction'])
    model.plot()
    print(f'h{i+1:.0f}: {model.h:.4f}')
    hsum += model.h
    if i == len(range(15))-1:
        print(f'Avg: {hsum/15:.4f}')
```

```
h1: 7.0643
h2: 6.6935
h3: 6.7948
h4: 6.8615
h5: 6.9025
h6: 6.6774
h7: 7.2006
h8: 7.0422
h9: 6.9545
h10: 7.3189
h11: 6.9008
h12: 6.7967
h13: 6.8160
h14: 7.0204
h15: 6.9235
Avg: 6.9312
```



C.

The most reliable region of the original plot will be between 20 and 45 because there is the greatest amount of data for this region. However, there is a lot of variation in the plots of the resampled data between 20 and 65, and very little outside of this range.